



# CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)  
REVIEW TEST  
08-08-2021

## JEE(Main + Advanced) : NURTURE COURSE [PHASE : S(TEPS)]

Time : 3 Hours

### PAPER-1

Maximum Marks : 180

#### READ THE INSTRUCTIONS CAREFULLY

#### GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

#### OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

#### DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take  **$g = 10 \text{ m/s}^2$**  unless otherwise stated.

#### QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

**SOME USEFUL CONSTANTS**

**Atomic No. :** H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,  
Cl = 17, Br = 35, Xe = 54, Ce = 58

**Atomic masses :** H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,  
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,  
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

**Space for Rough Work**

**HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS**

**BEWARE OF NEGATIVE MARKING**

**PART-1 : PHYSICS**

**SECTION-I (i) : (Maximum Marks: 24)**

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all ) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.

*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.

*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

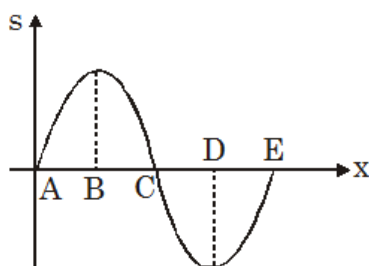
*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).

*Negative Marks* : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

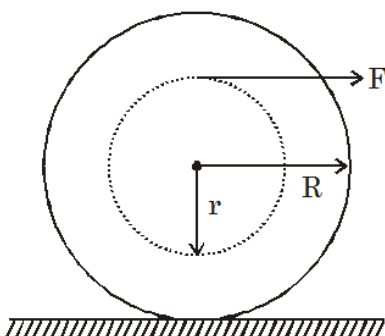
1. A sound wave is travelling along positive x-direction. Displacement(s) of particles from their mean position at any time  $t$  is shown in the figure. Choose the correct alternative(s) :



- (A) Particle located at E has its velocity in negative y-direction.
- (B) Particle located at D has zero velocity.
- (C) Particles located near C are under compression.
- (D) Change in pressure from equilibrium value at D is zero.

**Space for Rough Work**

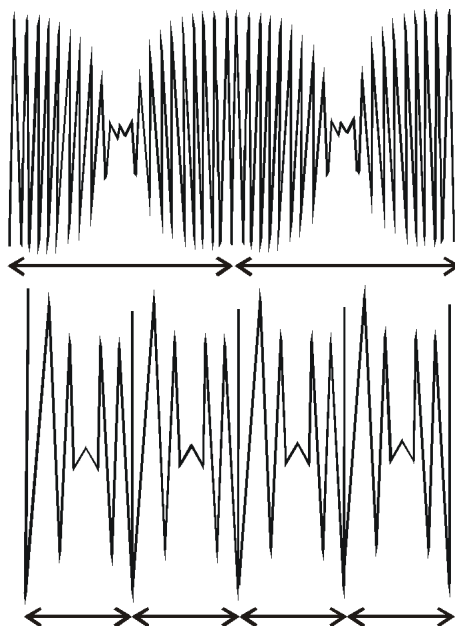
2. A string, of length  $L$ , clamped at both ends is vibrating in its first overtone mode. Select the correct option(s) from following for the moment the string looks flat.
- (A) The points whose speed is half the maximum speed may be at a distance of  $L/3$  from each other.
- (B) The points whose speed is half the maximum speed may be at a distance of  $L/2$  from each other.
- (C) Four points have speed one fourth the maximum speed.
- (D) Two points have maximum speed.
3. A heavy homogenous cylinder has mass  $m$  and radius  $R$ . It is accelerated by a force  $F$  which is applied to the cylinder at a distance  $r$  from the center. The coefficient of static friction is sufficient for the cylinder to roll without slipping then



- (A) The frictional force is  $\frac{2F}{3}$
- (B) The acceleration of the centre of the cylinder is  $\frac{2F}{3mR}(R + r)$
- (C) It is possible to choose  $r$  so that acceleration of centre of cylinder is greater than  $\frac{F}{m}$
- (D) The direction of frictional force is always opposite to  $F$

Space for Rough Work

4. Each of the two figures is rough illustration of the resulting waveform (y versus t) due to overlapping of two sound waves. The four component waves have frequencies of 300 Hz, 200Hz, 204 Hz and an unknown frequency slightly different from 300 Hz. Choose the correct statement(s) regarding this superposition pattern.

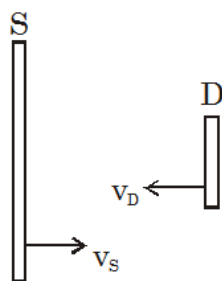


- (A) The figure at the top refers to the superposition of waves of frequency 300 Hz and the unknown frequency  
 (B) The figure at the top refers to the superposition of waves of frequency 200 Hz and 204 Hz  
 (C) The unknown frequency can be 298 Hz.  
 (D) The unknown frequency can be 306 Hz.

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**Space for Rough Work**

5. An elastic rubber rope of natural length 1 m and negligible mass hangs from the ceiling of the gymnasium. Ram hangs at the end of the rope and oscillates vertically with a time period of 1 seconds. Take :  $\pi^2 = 10$ . If the rope has a force constant of 2000 N/m,
- (A) Mass of Ram is 62.5 kg
- (B) The extension in the rope at the mean position is 25 cm
- (C) If ram were to hang at half the natural length of the string, the time period of oscillation would decrease
- (D) The time period of oscillation in option C is  $\frac{1}{\sqrt{2}}$  seconds
6. A large planar source 'S' oscillating at a frequency  $f$ , produces planar wavefronts. When source (S) and detector (D) are stationary in the orientation shown, the average rate of acoustic energy intercepted by the perfectly absorbing plane surface of the detector is  $P_0$ . Let velocity of sound be 'C'. If  $v_S = v_D = \frac{C}{5}$ , (direction of motion as shown in fig), then



- (A) Apparent frequency recorded by D is  $1.5 f_0$ .
- (B) Wavelength of the sound as received by D is  $(\lambda) = \frac{2C}{3f_0}$
- (C) The rate of acoustic energy intercepted by D is  $(P) = 1.5 P_0$
- (D) Beat frequency recorded by D is  $\frac{f_0}{2}$ .

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

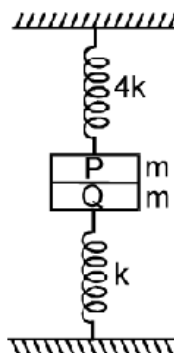
*Full Marks* : +3 If **ONLY** the correct option is chosen.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).

*Negative Marks* : -1 In all other cases.

Paragraph for Questions 7 and 8

In the figure shown, blocks P and Q are in contact but do not stick to each other. The lower face of P behaves as a plane mirror. The springs are in their natural lengths. The system is released from rest.

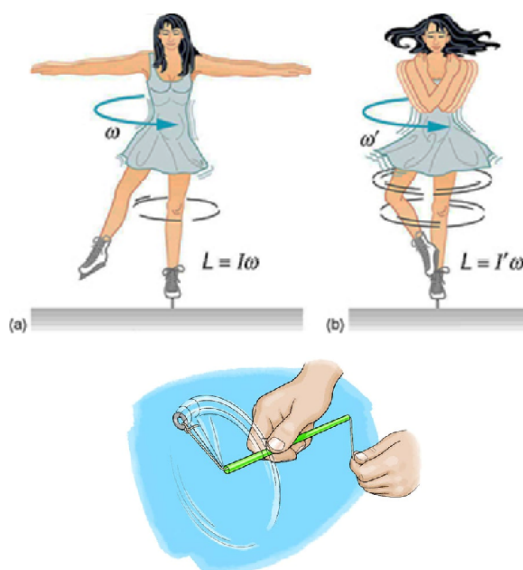


7. Then the distance between P and Q when Q is at the lowest point first time will be
- (A)  $\frac{2mg}{K}$                       (B)  $\frac{4mg}{K}$                       (C)  $\frac{3mg}{K}$                       (D) 0
8. In the above question, the maximum velocity of the image of Q in plane mirror of block P with respect to ground is at a time :
- (A)  $\frac{\pi}{3} \sqrt{\frac{m}{k}}$                       (B)  $\sqrt{\frac{m}{k}} \cos^{-1} \left( \frac{\sqrt{33} - 1}{8} \right)$
- (C)  $\sqrt{\frac{m}{k}} \cos^{-1} \left( \frac{\sqrt{33} + 1}{8} \right)$                       (D) None of these

Space for Rough Work

### Paragraph for Questions 9 and 10

An ice skater is spinning on the tip of her skate with her arms extended. In the next image, her rate of spin increases greatly when she pulls in her arms. A similar phenomenon can be observed with a washer being rotated as shown. For simplicity, assume no gravity and friction to be absent.



9. If the person holding the thread pulls the thread down so that the radius of the circular path of washer decreases,
- (A) The kinetic energy of washer increases, angular velocity increases and the angular momentum about the center remains constant.
  - (B) The kinetic energy of washer decreases, angular velocity increases and the angular momentum about the center remains constant.
  - (C) The kinetic energy of washer increases, angular velocity decreases and the angular momentum about the center remains constant.
  - (D) The kinetic energy of washer decreases, angular velocity decreases and the angular momentum about the center remains constant.
10. In the case of ice skater who pulls her hands and legs inwards,
- (A) Positive Work is done by the ice skater due to which her kinetic energy increases
  - (B) Negative work is done by the ice skater due to which her kinetic energy decreases
  - (C) Positive Work is done by the ice skater due to which her kinetic energy decreases
  - (D) Negative work is done by the ice skater due to which her kinetic energy increases

Space for Rough Work

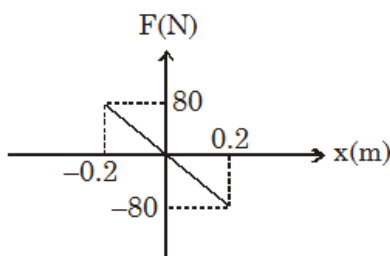
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

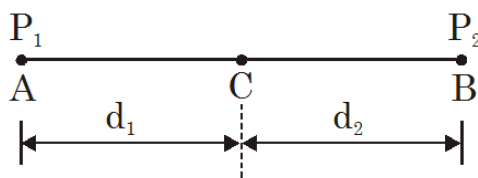
*Full Marks* : +3 If ONLY the correct numerical value is entered.

*Zero Marks* : 0 In all other cases.

1. A body of mass 0.1 kg executes simple harmonic motion (SHM) about  $x = 0$  under the influence of a force shown below. The time period of the SHM is (in sec) : Take :  $\pi^2 = 10$

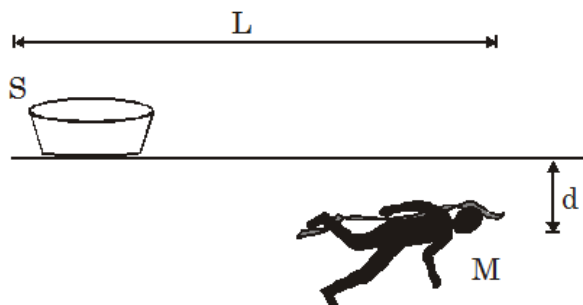


2. As shown in the figure, Two Loudspeakers are located at point A and B. Both are vibrating in phase at a frequency  $f$  and  $P_1$  and  $P_2$  are their respective power outputs. Point C lies on a line joining the two loudspeakers at a distance of  $d_1$  from A and  $d_2$  from B. With both speakers switched on what is the intensity (in  $\text{W/m}^2$ ) at point C. Take velocity of sound =  $300\text{ms}^{-1}$ , frequency  $f = 100\text{Hz}$ .  $d_1 = 1\text{ m}$  and  $d_2 = 1.5\text{ m}$ .  $P_1 = 8\pi$  watts and  $P_2 = 18\pi$  watts. Also, assume that loudspeakers behave like point isotropic sources, (emit sound uniformly in all directions).

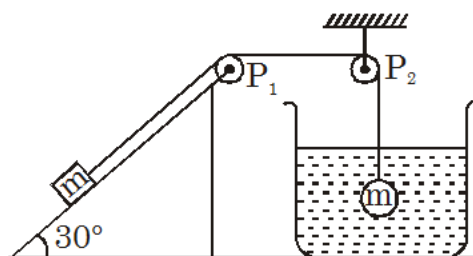


Space for Rough Work

3. A man is swimming at a depth  $d$  in a sea at a distance  $L$ , ( $L \gg d$ ) from a ship (S). An explosion occurs in the ship and after hearing the sound the man immediately moves to the surface. It takes  $0.8\text{s}$  for the man to rise to the surface after he hears the sound of explosion.  $0.2\text{s}$  after reaching the surface he once again hears a sound of explosion. Calculate  $L$  (in m). Given: Speed of sound in air =  $\frac{1000}{3} \text{ m/s}$ ; Bulk modulus of water =  $1 \times 10^9 \text{ Pa}$



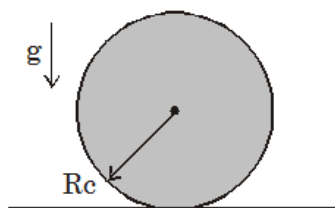
4. A block of mass ' $m$ ' is kept over a fixed smooth wedge. Block is attached to a sphere of same mass through fixed massless pulleys  $P_1$  and  $P_2$ . Sphere is inside the water as shown. If specific gravity of sphere is 2, find the acceleration of sphere. Neglect viscosity.



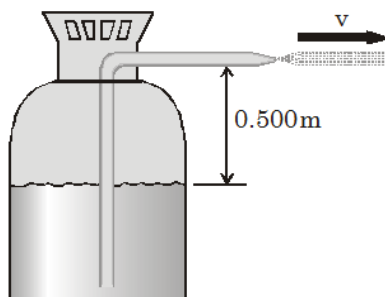
5. A source of sound is located in a medium in which speed of sound is  $V$  and an observer is located in a medium in which speed of sound is  $2V$ . Both the source and observer are moving directly towards each other with a velocity of  $V/5$ . If the frequency of sound is  $200 \text{ Hz}$ , what is the frequency (in Hz) detected by the observer?

Space for Rough Work

6. The distance between two identical high-rise buildings is  $L = 150$  m. Between the buildings a uniform wire is stretched so that near the attachment points they make up identical small angles  $\alpha = \frac{3^\circ}{\pi}$ . Climbing the roof, the schoolboy Ram conducts a dangerous experiment. He strikes a stick on the wire at the end and measures time in which pulse running through the wire returns to Ram after reflection from the other end of the wire. What is the time (in sec.) measured by Ram?
7. A liquid drop of radius  $R_c$  is placed on a smooth horizontal surface in uniform gravity  $g$ . Assume that the bulging shape of liquid drop due to gravity is negligible and it can be treated as a spherical drop. This liquid drop is made of  $N$  smaller identical liquid droplets. The density of liquid is  $K$  and surface tension is  $T$ . What should be the critical value of  $R_c$  (in cm) so that such type of drop can form by combining these drops without any energy input from outside? Take  $T = 0.3$  N/m,  $K = 1000$  kg/m<sup>3</sup>,  $N = 1000$ .



8. Water is forced out of a fire extinguisher by air pressure, as shown in Figure. How much gauge pressure (in  $10^5$  Pa) in the tank (above atmospheric) is required for the water jet to have a speed of  $30.0$  m/s when the water level in the tank is  $0.500$  m below the nozzle? Assume that the area of the nozzle is much smaller than that of the fire extinguisher.



Space for Rough Work

**PART-2 : CHEMISTRY**  
**SECTION-I (i) : (Maximum Marks: 24)**

- This section contains **SIX (06)** questions.
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- For each question, choose the option(s) corresponding to (all ) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.

*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.

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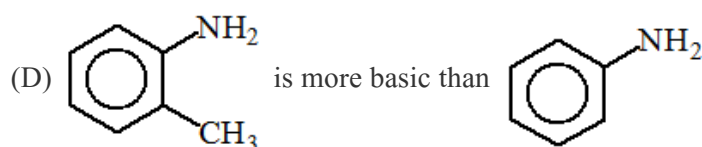
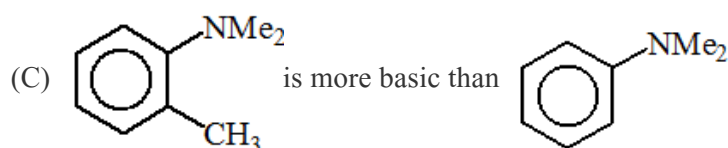
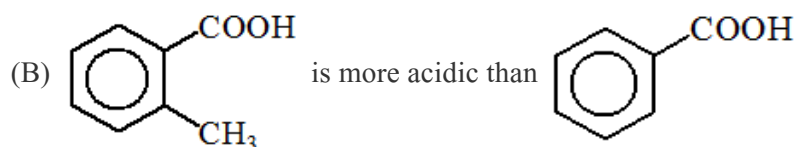
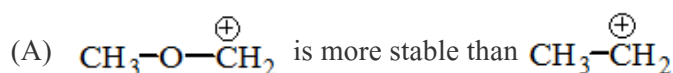
*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).

*Negative Marks* : -2 In all other cases.

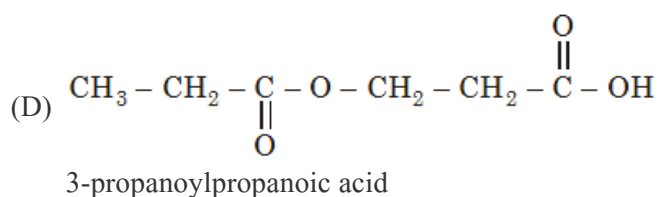
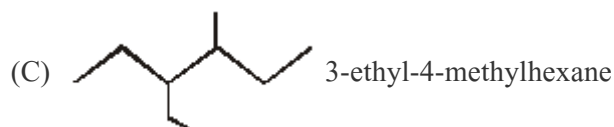
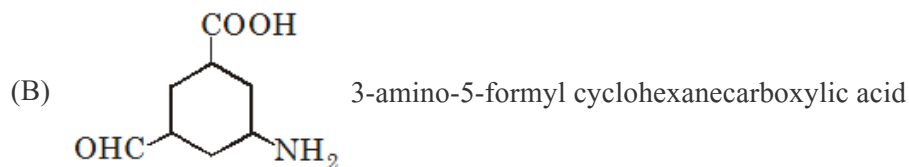
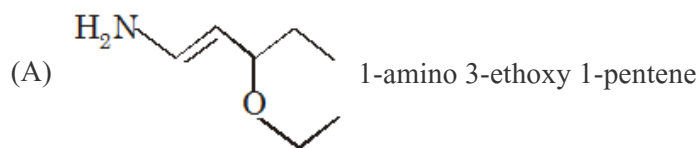
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Select correct statement among following -



Space for Rough Work

2. Which of following IUPAC names are correct:



3. NiO is doped with small quantity of  $\text{Li}_2\text{O}$ . This crystal

- (A) is a p-type semiconductor
- (B) is a n-type semiconductor
- (C) Will have vacant  $\text{O}^{2-}$  sites and a lower density as compared to pure NiO
- (D) Will have vacant  $\text{O}^{2-}$  sites and a higher density as compared to pure NiO

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Space for Rough Work

4. 100 g sample of clay (containing 19%  $\text{H}_2\text{O}$ , 40% silica, and inert impurities as rest) is partially dried so as to contain 10%  $\text{H}_2\text{O}$   
Which of the following is/are correct statements (s) ?
- (A) The percentage of silica in partially dried clay is 44.4%
- (B) The mass of partially dried clay is 90.0 g.
- (C) The percentage of inert impurity in partially dried clay is 45.6%
- (D) The mass of water evaporated is 10.0 g
5. Which of the following statement(s) is/are correct?
- (A)  $\text{p}K_1$  (maleic acid) <  $\text{p}K_1$  (fumaric acid)  
but  $\text{p}K_2$  (maleic acid) >  $\text{p}K_2$  (fumaric acid)
- (B) The solubility of noble gases in water increases with increase of size of inert gas
- (C) Urea is solid at room temperature while acetone is liquid
- (D) Boric acid is gas at room temperature like  $\text{BF}_3$
6. Select the correct order(s) for the mentioned property?
- (A)  $\text{Fe}^{+2} < \text{Fe}^{+3} < \text{Fe}^{+4}$  (Spin only magnetic moment)
- (B)  $\text{Al} < \text{Ga} < \text{Tl} < \text{B}$  (Ionisation energy)
- (C)  $\text{O}^+ < \text{Te}^+ < \text{Se}^+ < \text{S}^+$  (Electron affinity)
- (D) Mendeleev's periodic table < Modified Mendeleev Periodic table < Long form of periodic table (number of groups)

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Space for Rough Work

**SECTION-I (ii) : (Maximum Marks: 12)**

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
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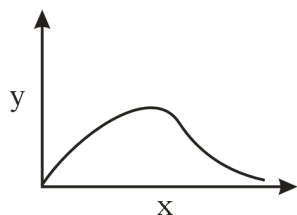
*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).

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**Paragraph for Questions 7 and 8**

Ervin Schrodinger developed a model which is based on the particle and wave nature of the electron, known as Wave Mechanical Model of atom. The equation determines the behaviour of the wave function that describes the wave like properties of subatomic system. It is solved to find the different energy levels of the system.

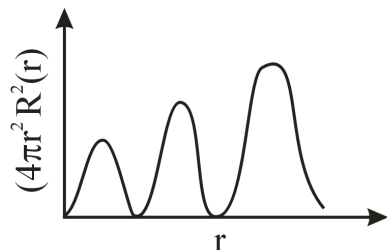
7. Match the nature of this graph with appropriate x & y axes ?



- (A)  $x = (r)$ ,  $(y) = \Psi_{200}(r)$
- (B)  $x = (r)$ ,  $(y) =$  Radial probability density of 2s
- (C)  $x = (r)$ ,  $(y) =$  Radial probability of  $\Psi_{311}$
- (D)  $x = (r)$ ,  $(y) =$  Radial wave function of  $n = 2$ ,  $l = 1$ ,  $m = 0$

**Space for Rough Work**

8. For similar nature of below radial probability graph where  $n \leq 5$ , select incorrect statement



- (A) Angular node present in orbital may be 1  
 (B) Angular nodes present in orbital may be 2  
 (C) For possible orbitals magnetic quantum number may be 2  
 (D) For possible value of total nodes must be less than 4

### Paragraph for Questions 9 and 10

Those acidic compounds which contains X-O-H bond are called oxy, where X is usually a non metal.

9. Select the di basic {-ic} acid in which atleast one oxy linkage is present.  
 (A)  $\text{H}_4\text{P}_2\text{O}_5$                       (B)  $\text{H}_2\text{S}_2\text{O}_7$                       (C)  $\text{H}_3\text{P}_3\text{O}_9$                       (D)  $\text{H}_2\text{S}_2\text{O}_3$
10. Select the molecule in which two tetrahedral units are joined together by common vertex?  
 (A) Marshall's acid                      (B) Pyrosulphurous acid  
 (C) Pyrosulphuric acid                      (D) Dithionic acid

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Space for Rough Work

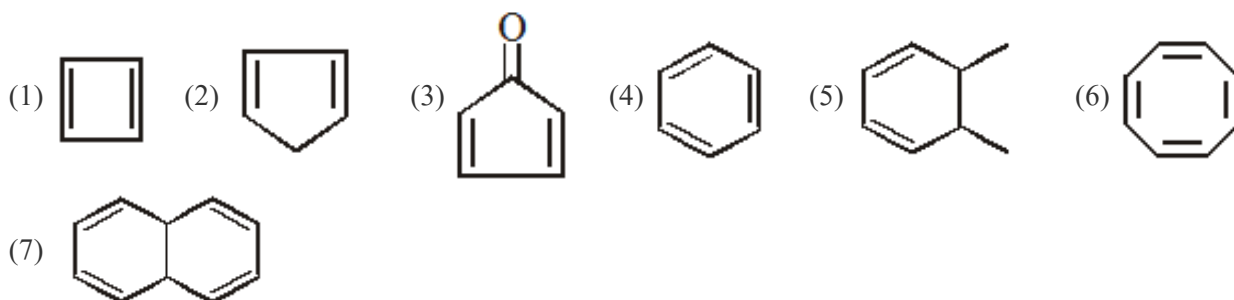
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +3 If ONLY the correct numerical value is entered.

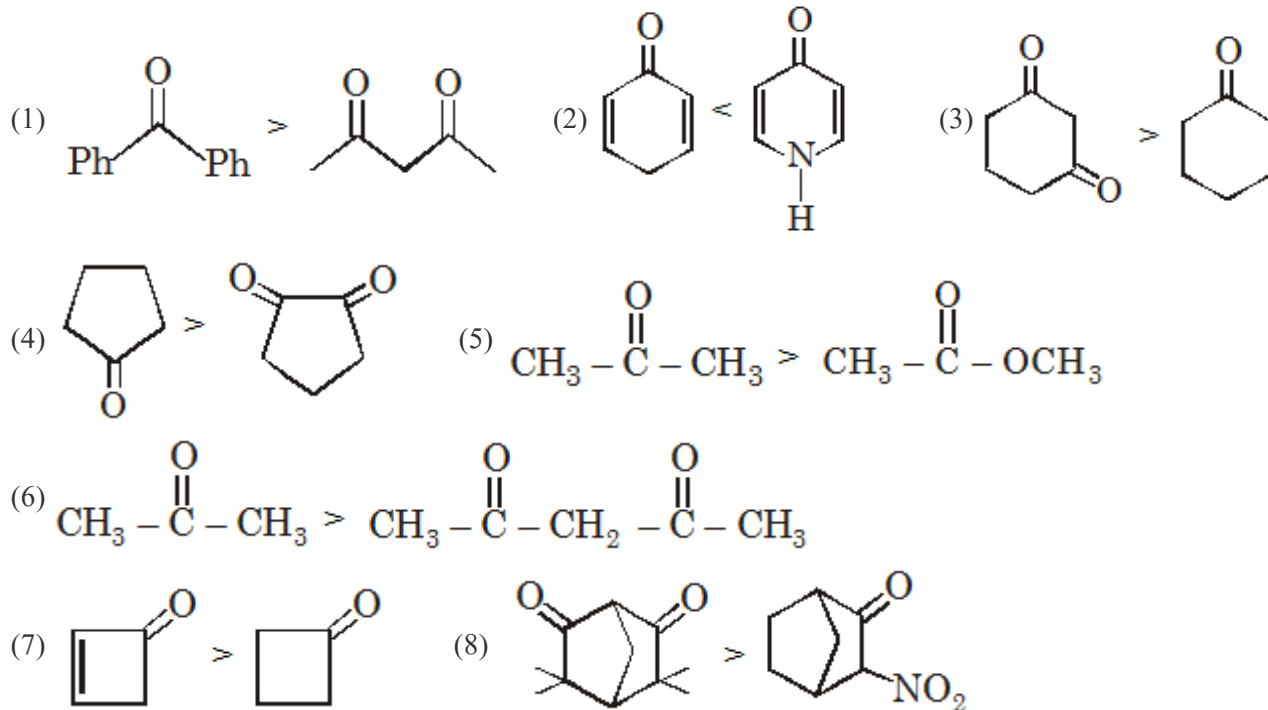
*Zero Marks* : 0 In all other cases.

1. Identify total number of compounds out of following which are unstable at room temperature?



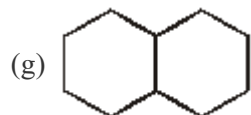
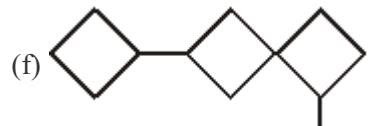
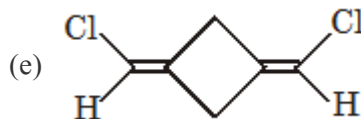
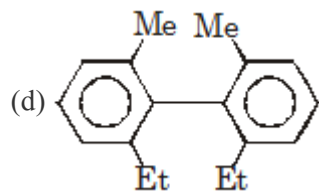
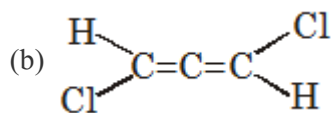
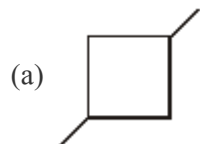
Space for Rough Work

2. How many of the following pairs showing correct order of percentage enolic content.

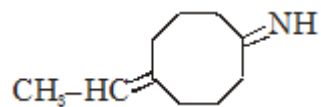


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3. How many compounds show geometrical isomerism ?



4. Total number of Geometrical isomer of following compound is :



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5. A mineral of iron contains an oxide containing 72.41 % iron by mass and has a density of 6.04 g/cc. Its unit cell is cubic with edge length of 800 pm. What is the total number of atoms (ions) present in each unit cell ? (Fe -56, O-16)
6.  $\text{COF}_2(\text{g})$  passed over catalyst at  $1000^\circ\text{C}$  comes to equilibrium :  $2\text{COF}_2(\text{g}) \rightleftharpoons \text{CO}_2(\text{g}) + \text{CF}_4(\text{g})$ . Analysis of the equilibrium mixture (after quick cooling to freeze the equilibrium) shows that 500 ml of the equilibrium mixture (STP) contains 300 ml (STP) of  $(\text{COF}_2 + \text{CO}_2)$  taking the total pressure to be 10 atm. The value of  $K_p$  for the reaction is
7. Find the sum of the species from the following in which all bond lengths are not equal?  
 $\text{ClF}_3$ ,  $[\text{SF}_5]^+$ ,  $\text{ClF}_5$ ,  $[\text{XeF}_3]^+$ ,  $\text{O}_3$ ,  $\text{P}_4$  (white),  
 Fullerene $_{(\text{C}_{60})}$ ,  $\text{Si}_2\text{O}_7^{2-}$ ,  $\text{P}_4\text{O}_6$ ,  $\text{P}_4\text{O}_{10}$
8. Consider the following compounds  $\text{N}_2\text{O}_{4(\text{s})}$ ,  $\text{PCl}_{5(\text{s})}$ ,  $\text{I}_2\text{Cl}_{6(\text{l})}$ ,  $\text{N}_2\text{O}_{5(\text{s})}$ ,  $\text{XeF}_{6(\text{s})}$   
 Calculate (x + y)  
 x = sum of different type of multiatomic ions which are planar in shape  
 y = sum of type of ions in which  $3d_{z^2}$  orbital is involved in hybridisation

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**Space for Rough Work**

**PART-3 : MATHEMATICS**  
**SECTION-I (i) : (Maximum Marks: 24)**

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all ) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.

*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.

*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).

*Negative Marks* : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Let AB be a focal chord (not the latus rectum) of the parabola  $y^2 = 4px$  where  $p$  is a prime number ( $p > 0$ ), such that the lengths SA and SB are integers (S being focus of the parabola). If the tangents to the parabola at A and B meet at point P, then

(A)  $\frac{PA + PB - |PA - PB|}{2} = (p + 1)\sqrt{p + 1}$

(B)  $\frac{PA + PB + |PA - PB|}{2} = (p + 1)\sqrt{(p + 1)p}$

(C)  $SP = (p + 1)\sqrt{p}$

(D)  $AB = (p + 1)^2$

**Space for Rough Work**

2. Let  $f(x) = x^{2018} + a_0x^{2017} + a_1x^{2016} + a_2x^{2015} + \dots + a_{2017}$  where  $a_0, a_1, a_2, \dots, a_{2017}$  are real numbers such that  $f(0) = f(1)$  and each root of  $f(x) = 0$  is real and always lies between 0 and 1 then product of roots is
- (A) less than or equal to  $\frac{1}{2^{2018}}$  (B) less than or equal to  $\frac{1}{2^{2017}}$
- (C) greater than or equal to  $\frac{1}{2^{2018}}$  (D) greater than  $\frac{1}{2^{2019}}$
3. Choose the correct statement(s) -
- (A) Each focal chord subtend obtuse angle at the vertex of the parabola.
- (B) An ellipse whose one of the focus being fixed at a point P, is sliding along two fixed straight lines  $L_1$  &  $L_2$  (P does not lie on any of  $L_1$  or  $L_2$ ), then locus of its center is a circle.
- (C) A triangle of 8 sq. unit area is inscribed in parabola  $y^2 = 4x$ . Then sum of absolute difference of ordinates of vertices, taken pairwise, is always greater than 12.
- (D) A pair of tangents are drawn to  $x^2 - \frac{y^2}{4} = 1$  from a point  $\left(2, \frac{7}{2}\right)$ , then both points of contact will lie in Ist quadrant.

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Space for Rough Work

4. Consider a rectangle with vertices  $A \equiv (0, 0)$ ,  $B \equiv (4, 0)$ ,  $C \equiv (4, 6)$  and  $D \equiv (0, 6)$ . Let  $S$  be the region consisting of all points  $P$  inside the rectangle such that  $d(P, DC) < d(P, AB) < \max.\{d(P, BC); d(P, AD)\}$ , where  $d(P, MN)$  denotes distance of the point  $P$  from the line joining points  $M$  and  $N$ , then
- (A)  $S$  is symmetrical about the line  $y = 3$   
 (B)  $S$  is symmetrical about the line  $x = 2$   
 (C) Area of  $S$  is 2 sq. units  
 (D) Area of  $S$  is 1 sq. unit
5. For all real number  $x$ ,  $2^x + 3^x - 4^x + 6^x - 9^x$  is always less than -
- (A) 0 (B) 1 (C) 2 (D) 3
6. If  $(x + 1)(x + 2) \dots (x + 99) = A_0 + A_1 \cdot x + A_2 \cdot x^2 + \dots + A_{99} \cdot x^{99}$  and  $S = \frac{\sum_{k=1}^{99} (k \cdot A_k)}{100!} = \sum_{i=1}^{99} \left( \frac{1}{r_i} \right)$ ,  $(r_i < r_j, \forall i < j)$  then value of  $\sum_{k=1}^{99} \frac{1}{\sqrt{(r_k - 1)}}$  is lying between
- (A) 15 and 19 (B) 15 and 17 (C) 14 and 17 (D) 17 and 20

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**Space for Rough Work**

**SECTION-I (ii) : (Maximum Marks: 12)**

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +3 If **ONLY** the correct option is chosen.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).

*Negative Marks* : -1 In all other cases.

**Paragraph for Questions 7 and 8**

Consider ellipse  $E : Ax^2 + By^2 + Hxy - 2gx - 2fy + 64 = 0$  touches line  $y = 0$ . Let  $(3, 2)$  is centre of ellipse and  $(5, 1)$  is one focus of ellipse.

7. Eccentricity of ellipse is

(A)  $\sqrt{\frac{5}{8}}$  (B)  $\sqrt{\frac{4}{7}}$  (C)  $\sqrt{\frac{5}{7}}$  (D)  $\sqrt{\frac{7}{8}}$

8. Value of  $A + B + H$  is

(A) 16 (B) 12 (C) 14 (D) 15

**Paragraph for Questions 9 and 10**

Consider system of equation in variables  $\alpha, \beta$ ,

$[\sin(\alpha - \beta) + \cos(\alpha + 2\beta)\sin\beta]^2 = 4\cos\alpha\sin\beta\sin(\alpha + \beta)$  and  $\tan\alpha = 3\tan\beta$ . (Wherever defined)

9. If sum of all possible values of  $\sin^2\beta$  is  $\frac{a}{b}$  (where  $a$  and  $b$  are coprime numbers), then value of  $2a + b$  is -

(A) 17 (B) 25 (C) 13 (D) 22

10. Number of possible value(s) of  $\beta$  in  $\beta \in [-\pi, 4\pi]$ , is -

(A) 10 (B) 12 (C) 14 (D) 16

**Space for Rough Work**

**SECTION-II : (Maximum Marks: 24)**

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +3 If ONLY the correct numerical value is entered.

*Zero Marks* : 0 In all other cases.

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1. Line  $8x - 3y - 12 = 0$  cuts the parabola  $y^2 + 4 = 4x$  in points P & Q and its directrix in point E. QQ' is a focal chord and Q'P meets the directrix in point E'. If the area of triangle ESE' is A, then value of [A] is (where [.] denotes greatest integer function and S is focus of parabola)
  2. Given, the equation  $x^3 - 75x + a = 0$  has three integral roots  $\alpha$ ,  $\beta$  and  $\gamma$  (where  $a \in \mathbb{I}$ ) then  $\left( \frac{|\alpha| + |\beta| + |\gamma|}{4} \right)$  is equal to
  3. Let tangents be drawn to the curve  $y^2 - 4(x + y) = 3 \sin \theta + 4 \cos \theta - 15$ ,  $\theta \in \mathbb{R}$  from the origin where the vertex of the curve is at maximum distance from the origin. If slopes of these tangents are  $m_1$  and  $m_2$ , then find the value of  $|m_1 m_2|$ .
  4. Consider a branch of the hyperbola  $y^2 - 2x^2 - 2\sqrt{2}y + 4\sqrt{2}x - 6 = 0$  with vertex at the point A. Let B and C be end points of latus rectum of the hyperbola nearest to the point A. If area of triangle ABC is  $\sqrt{p} - q$ , where  $p, q \in \mathbb{N}$  then value of  $(p + q)$  is
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**Space for Rough Work**

5. If A, B, C are angles of an acute angle triangle. The minimum value of

$$\frac{1}{3^5} \begin{vmatrix} (\tan B + \tan C)^2 & \tan^2 A & \tan^2 A \\ \tan^2 B & (\tan C + \tan A)^2 & \tan^2 B \\ \tan^2 C & \tan^2 C & (\tan A + \tan B)^2 \end{vmatrix}$$

is

6. A straight line drawn through the point  $P(-1,2)$  meets the hyperbola  $xy = c^2$  at the points A & B (Point A & B lie on same side of P). Let Point Q is collinear with A & B such that PA, PQ & PB are in A.P. If locus of Q is  $y = \frac{\lambda_1 x}{(1 + \lambda_2 x)}$ , then  $\lambda_1 + \lambda_2$  is equal to
7. From point P on hyperbola  $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$  in first quadrant, line parallel to x-axis & y-axis are drawn cutting hyperbola at Q & R respectively. Let A be the vertex of hyperbola in 1<sup>st</sup> quadrant. If Area  $(\Delta APQ) = 2$  Area  $(\Delta APR)$  then eccentric angle of P is  $\frac{a\pi}{b}$  (where a & b are relatively prime integers). The value of a + b is
8. Consider a family of lines  $(4b + 3)x - (b + 1)y = 2b + 1$ , where  $b \in \mathbb{R}$ . The minimum area of the triangle which a member of this family with negative gradient makes with the positive semi-axes is equal to

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Space for Rough Work

NAME OF THE CANDIDATE .....

FORM NO. ....

I have read all the instructions  
and shall abide by them.

\_\_\_\_\_  
Signature of the Candidate

I have verified the identity, name and Form  
number of the candidate, and that question  
paper and ORS codes are the same.

\_\_\_\_\_  
Signature of the Invigilator

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